9 EXAM PRACTICE

1 Look at the following statements. Which statement explains correctly why a small increase in temperature leads to a significant increase in the rate of a gas phase reaction?

A The frequency of collisions between molecules is greater at a higher temperature.

B The activation energy of the reaction is less when the gases are at higher temperature.

C The frequency of collisions between molecules with kinetic energy greater than the activation energy is greater at a higher temperature.

D The average kinetic energy of the molecules is slightly greater at a higher temperature. [1]

(Total for Question 1 = 1 mark)

2 An exothermic reaction proceeds by two stages.

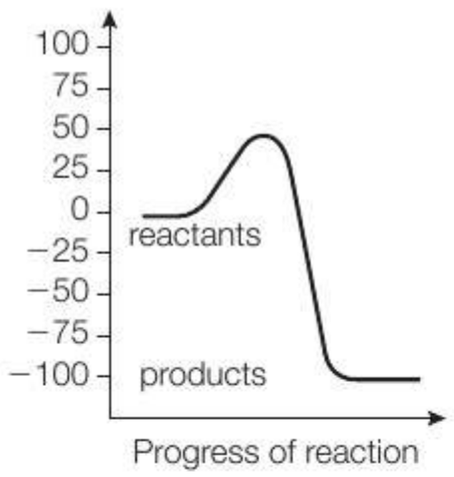
reactants $\xrightarrow{\text{stage 1}}$ intermediate $\xrightarrow{\text{stage 2}}$ products

The activation energy of stage 1 is $50\,kJ\,mol^{-1}$.

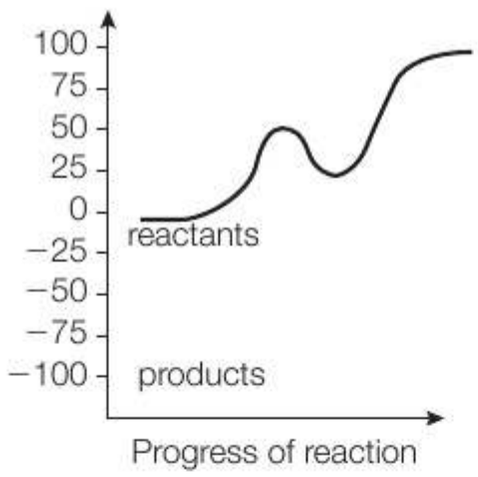
The overall enthalpy change of reaction is $-100\,kJ\,mol^{-1}$.

Which diagram could represent the energy profile diagram for the reaction?

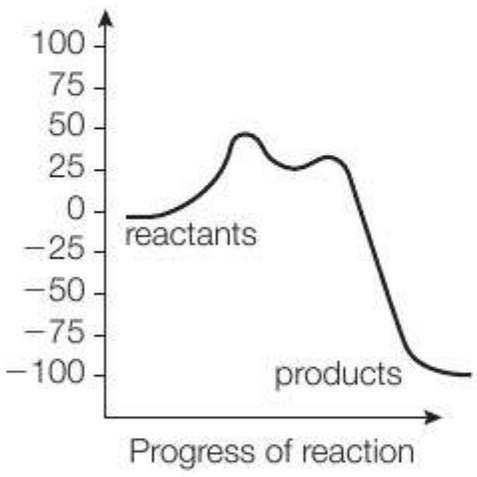
A



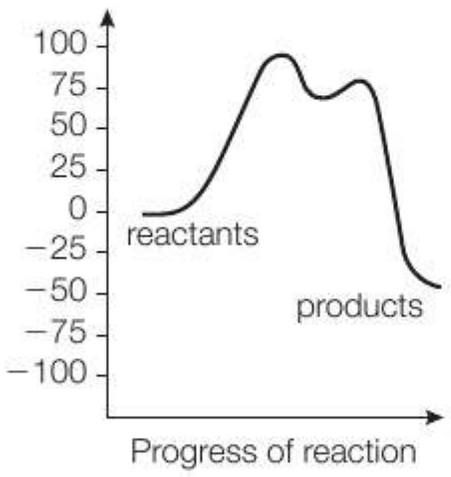
B



C



D

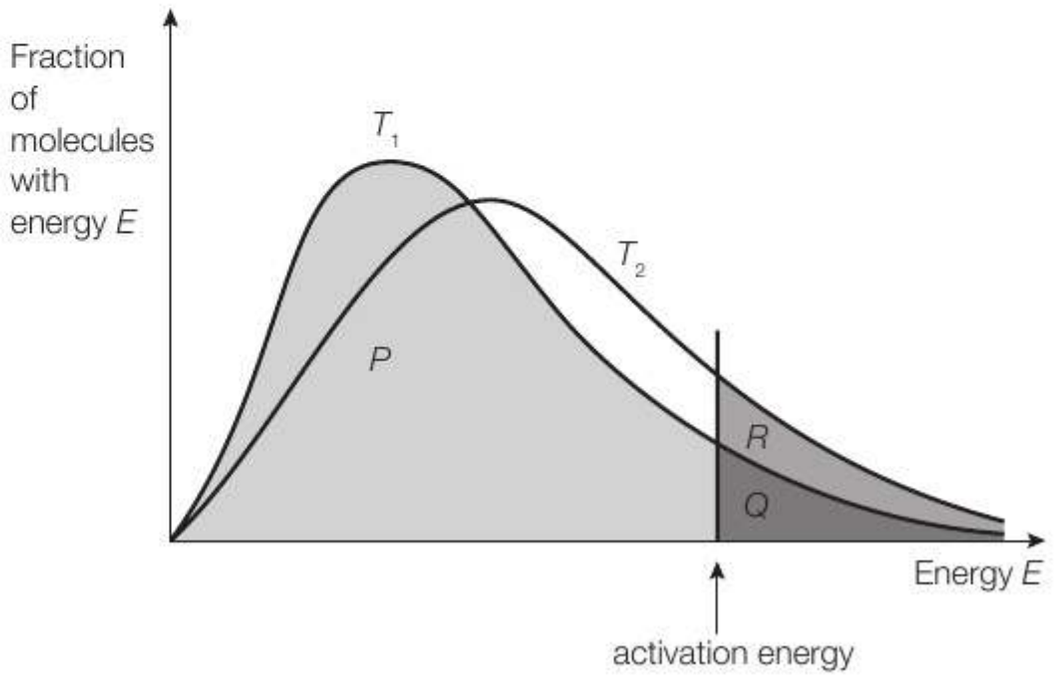


[1]

(Total for Question 2 = 1 mark)

3 The distribution of the fraction of molecules with energy E is given in the diagram for two different temperatures T_1 and T_2. T_2 is greater than T_1.

The letters P, Q and R represent separate and different shaded areas under the curves.

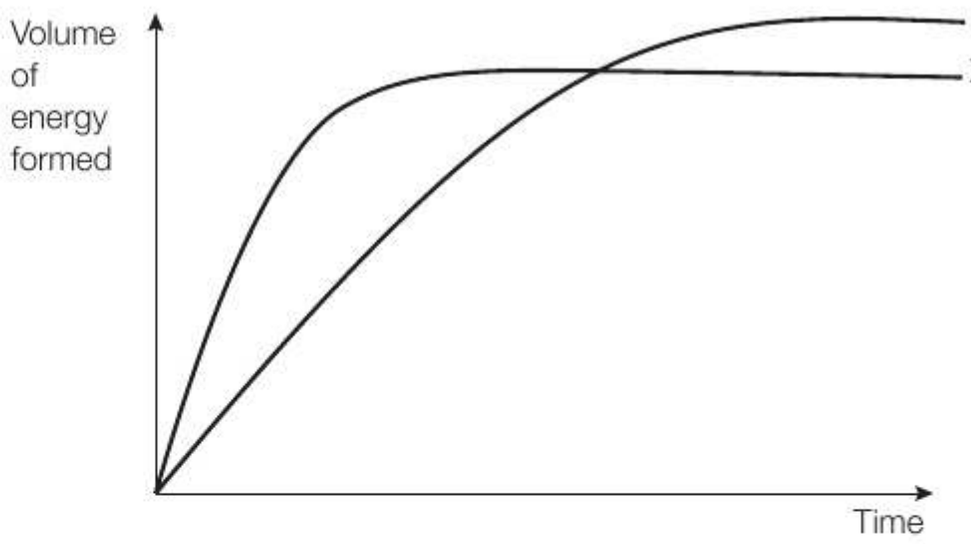


Which expression gives the fraction of molecules that have energy equal to or greater than the marked activation energy?

A $\frac{Q}{P}$ **B** $\frac{Q+P}{P}$ **C** $\frac{Q+R}{P+Q}$ **D** $\frac{Q+R}{P+Q-R}$ [1]

(Total for Question 3 = 1 mark)

4 Curve **X** in the diagram shows the volume of oxygen given off during the decomposition of $100\,cm^3$ of $1.0\,mol\,dm^{-3}$ hydrogen peroxide, catalysed by manganese(IV) oxide.



Which of the following alterations to the original experimental conditions would produce Curve **Y**?

A adding water

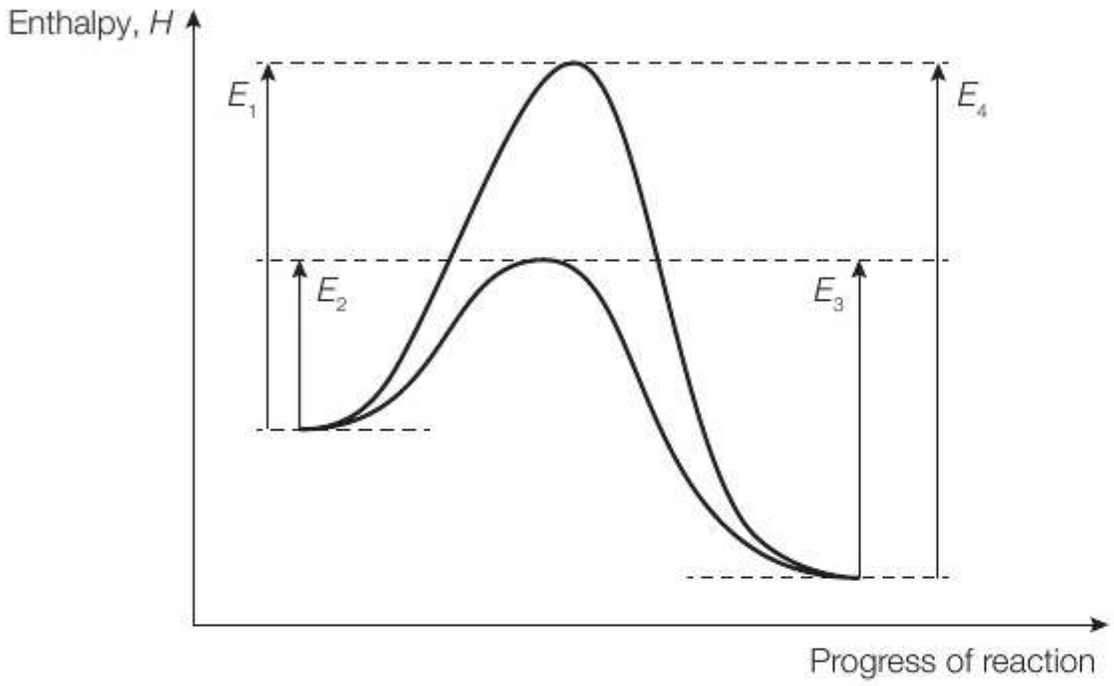
B adding some $0.1\,mol\,dm^{-3}$ hydrogen peroxide

C using less manganese(IV) oxide

D lowering the temperature [1]

(Total for Question 4 = 1 mark)

5 The enthalpy level diagram represents a reaction occurring with and without a catalyst.



Which of the following statements is correct?

A E_4 is the activation energy for the reverses reaction.

B The forward reaction, with catalyst, is endothermic.

C The enthalpy change of reaction is $(E_2 - E_3)$.

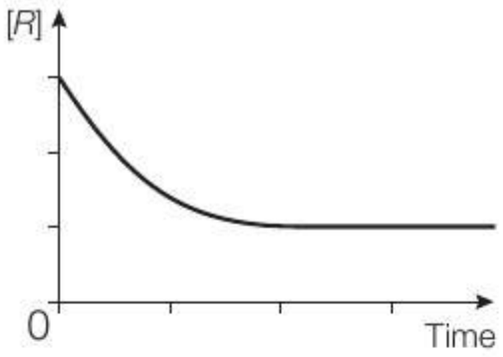
D The enthalpy change of reaction is reduced using a catalyst. [1]

(Total for Question 5 = 1 mark)

6 (a) *R* reacts with *T* in a reversible, exothermic reaction. The reaction is allowed to reach equilibrium.

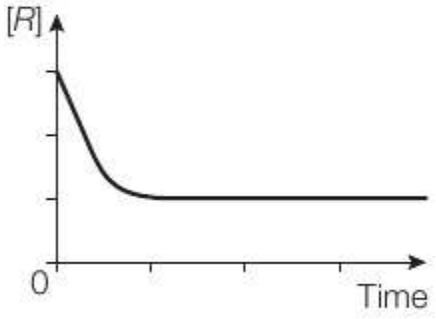
The concentration of one reactant, *R*, is plotted against time.

The graph shows the results.

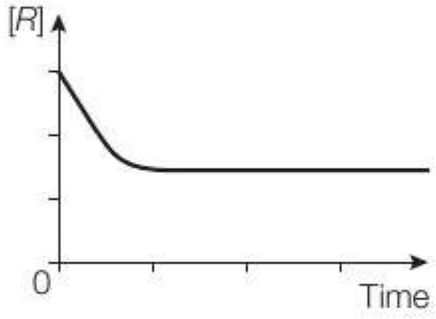


Which of the graphs below could be obtained if the reaction were repeated at a higher temperature, but with the same concentrations of *R* and *T*?

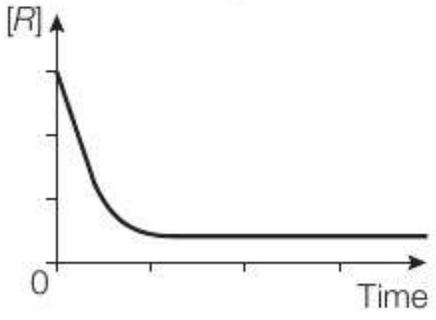
A



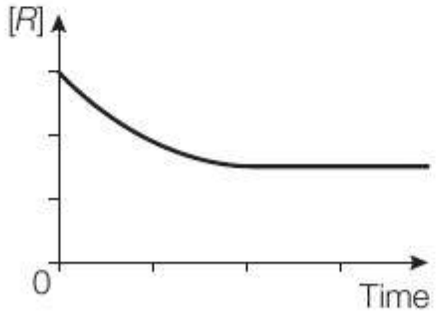
B



C



D



[1]

(b) Which statement about the effect of a catalyst on an endothermic, reversible reaction that is in dynamic equilibrium is correct?

A It increases decreases the yield of the product.

B It increases the rate of the forward reaction but not that of the backward reaction.

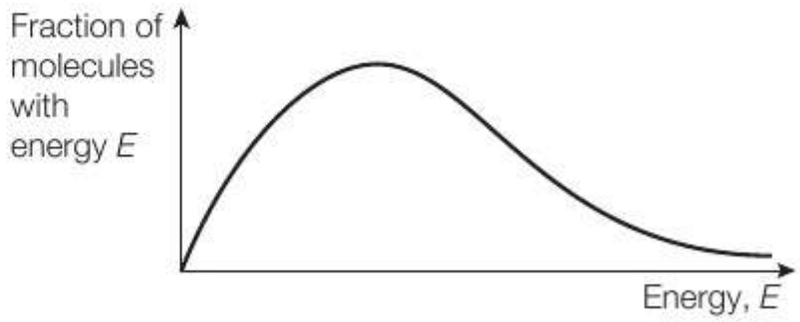
C It increases the rate of the backward reaction but not that of the forward reaction.

D It increases the rate of both the forward and the backward reactions. [1]

(Total for Question 6 = 2 marks)

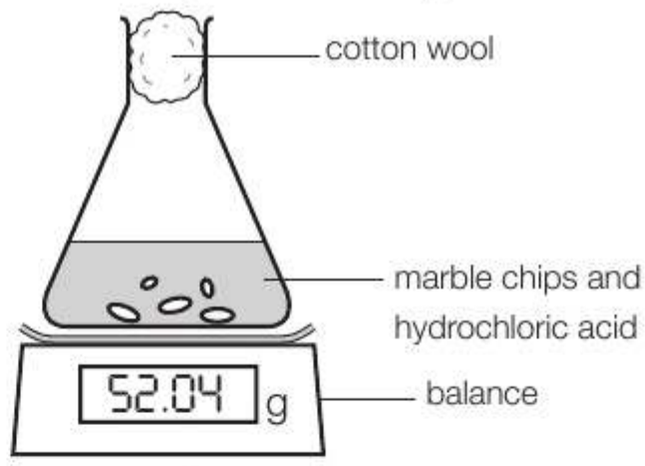
7 This question is about some graphs that you may have seen during your course.

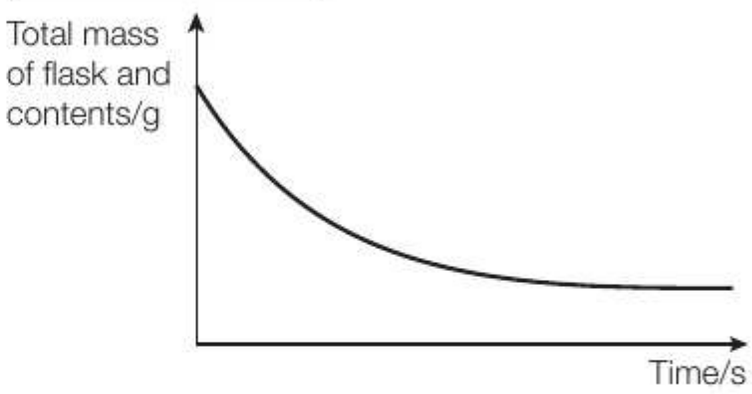
(a) The graph below is a Maxwell-Boltzmann distribution curve.



Explain, using the graph, the effect of adding a catalyst on the rate of reaction. [4]

(b) The graph below shows the change in mass observed in an experiment to investigate the rate of reaction between marble chips and dilute hydrochloric acid.





(i) State the purpose of the cotton wool in the neck of the conical flask. [1]

(ii) Explain why the mass of the flask and contents decreases during the course of the experiment. [2]

(iii) The experiment was repeated at the same temperature, with the same volume and concentration of acid and with the same mass of powdered marble.

Sketch, on the graph, the curve you would expect to obtain. [2]

(Total for Question 7 = 9 marks)

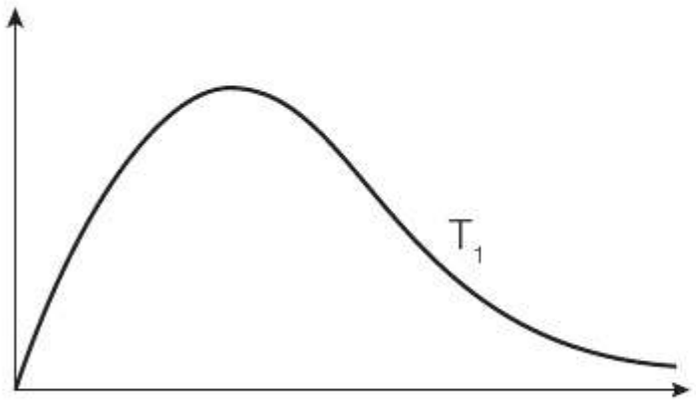
8 This question is about the effect that an increase in temperature has on the rate of a chemical reaction.

(a) A student found this statement in a text book:

'The rate of a chemical reaction increases as the temperature increases because the particles collide more often.'

Discuss the extent to which this statement is true. [5]

(b) The graph below shows a Maxwell-Boltzmann distribution curve for a gas at temperature T_1.



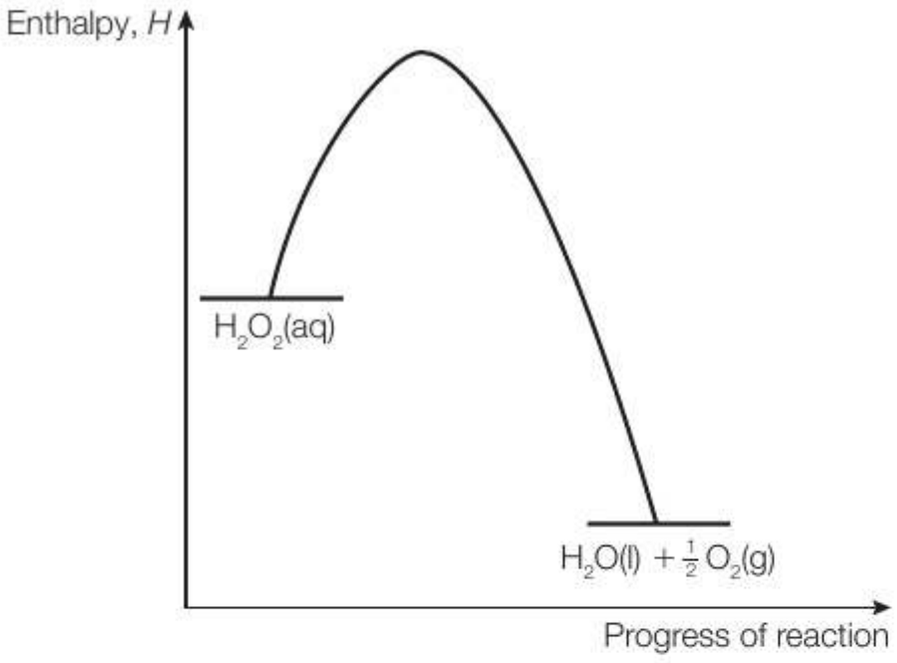
(i) Label the axes on the diagram. [2]

(ii) On the graph, sketch a curve to represent the distribution at higher temperature.

Label this curve T_2. [2]

(Total for Question 8 = 9 marks)

9 The graph below shows the reaction profile for the uncatalysed decomposition of hydrogen peroxide (H_2O_2) into water and oxygen.



(a) On the graph, label the activation energy (E_a) and the enthalpy change of reaction (ΔH). [2]

(b) The decomposition is catalysed by the addition of solid manganese(IV) oxide (MnO_2).

(i) Draw the curve to represent the catalysed decomposition. [2]

(ii) State, in principle, how the catalyst works in this reaction. [2]

(c) State whether the reaction is exothermic or endothermic. Justify your answer. [1]

(Total for Question 9 = 7 marks)

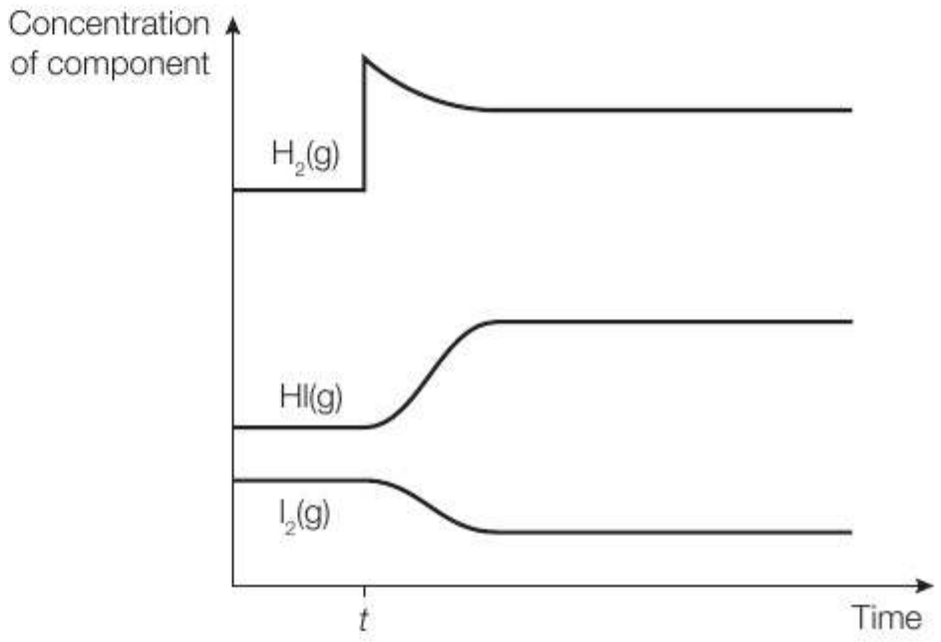
10 Many chemical reactions are easily reversible and can form equilibrium mixtures. Hydrogen and iodine react together in the gaseous state to form hydrogen iodide:

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g) \qquad \Delta_r H^\ominus = +52\,kJ\,mol^{-1}$$

Hydrogen (colourless) and iodine vapour (purple) are placed into a sealed container at 400 °C and allowed to reach equilibrium with hydrogen iodide (colourless). The resulting mixture has a very pale purple colour.

(a) When the reaction mixture is cooled to 200 °C, the reaction mixture becomes notably darker in colour. Give a reason for this observation in terms of the change to the equilibrium composition of the mixture. [2]

(b) In another experiment, all three gases were mixed together in a sealed tube and once again allowed to establish an equilibrium. The concentrations of the gases were monitored and the results were plotted on a graph. At time *t*, a change was made to the composition of the mixture.



(i) State the change that was made to the reaction mixture at time *t*. [1]

(ii) Describe the changes that occur to the reaction mixture after time *t*. [3]

(Total for Question 10 = 6 marks)

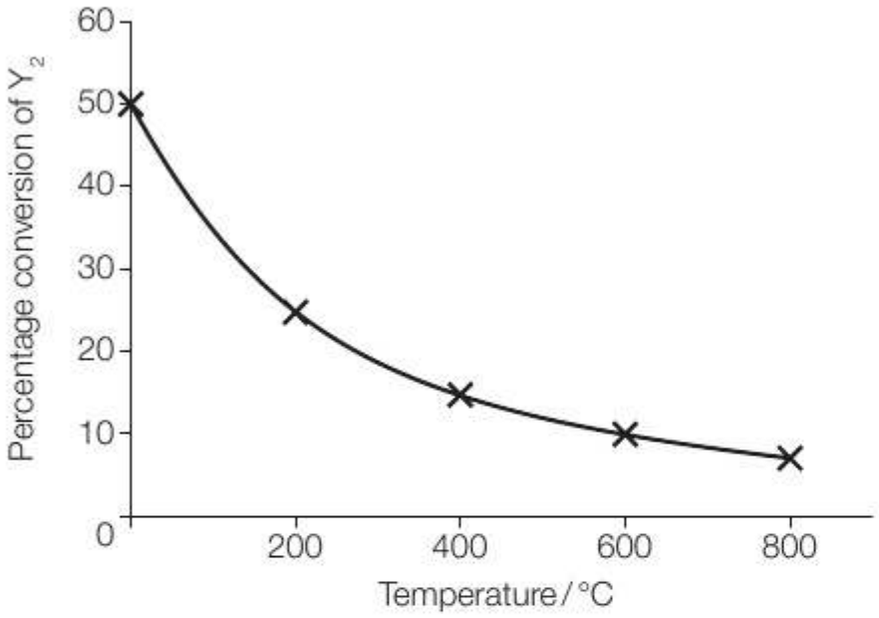
11 Many industrial processes involve reversible reactions. Chemists were investigating the production of a chemical XY_2 that can be formed from X_2 and Y_2 as shown in the equation

$$X_2(g) + 2Y_2(g) \rightleftharpoons 2XY_2(g)$$

The chemists carried out a series of experiments at different temperatures, each time allowing the reaction mixture to establish equilibrium.

The chemists measured the percentage conversion of Y_2 at each temperature.

The results are shown in the graph.



(a) Explain how the graph shows that the forward reaction is exothermic. [2]

(b) The chemists decide to use a catalyst in the process. Explain the effect of a catalyst on:

(i) the rate at which equilibrium is established

(ii) the percentage conversion of Y_2 at equilibrium. [4]

(Total for Question 11 = 6 marks)

12 Methanol (CH_3OH) can be manufactured from carbon monoxide and hydrogen.

$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g) \quad \Delta_r H^{-} = -128\,kJ\,mol^{-1}$$

(a) Predict how the composition of the equilibrium mixture is affected by:

(i) carrying out the reaction at a higher temperature but keeping the pressure the same

(ii) carrying out the reaction at a higher pressure but keeping the temperature the same.

In each case justify your answer. [4]

(b) Assuming the reaction takes place by collision of molecules in the gas phase, explain the effect on the rate of reaction of increasing the pressure. [2]

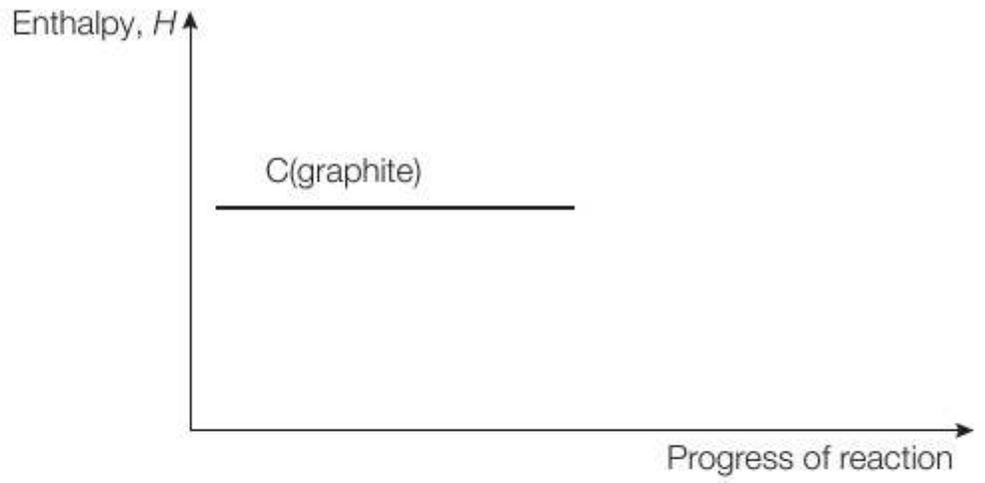
(Total for Question 12 = 6 marks)

13 Graphite and diamond are two allotropes of carbon. Artificial diamonds can be made from graphite.

$$C(graphite) \rightleftharpoons C(diamond) \quad \Delta_r H^{\ominus} = +1.8\,kJ\,mol^{-1}$$

Substance	**Density / g cm^{-3}**
graphite	2.25
diamond	3.51

(a) Complete the enthalpy profile diagram below for the conversion of graphite into diamond. Label the enthalpy change, ΔH. [2]



(b) State and justify which allotrope is the more thermodynamically stable. [1]

(c) State and justify in which allotrope the atoms take up less space per mole of substance. [1]

(d) Predict the conditions of temperature and pressure that are required to convert graphite into diamond. Justify your predictions. [4]

(Total for Question 13 = 8 marks)

14 An acid-base indicator, HIn, dissolves in water to form a green solution. The following equilibrium is established:

$$HIn(aq) \rightleftharpoons H^+(aq) + In^-(aq)$$
(yellow) (blue)

The indicator appears green because it contains sufficient amounts of both the yellow and blue species.

(a) Hydrochloric acid is added to the indicator solution. State what you would observe. Give a reason for your answer. [2]

(b) Aqueous sodium hydroxide is added dropwise to the resulting solution from (a) until no further colour change takes place. State all of the colours that would be observed. Give reasons for your answers. [4]

(Total for Question 14 = 6 marks)

15 The hydrogen required for the manufacture of ammonia and margarine is made by reacting methane with steam. The two gases react in a reversible reaction as shown by the equation:

$$CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g) \quad \Delta_r H^{\ominus} = +210\,kJ\,mol^{-1}$$

(a) State the conditions of temperature and pressure that would provide a high yield of hydrogen. Justify your answers. [4]

(b) State the conditions of temperature and pressure that would produce a fast rate of reaction. Justify your answers. Assume that the reaction takes place by molecules colliding in the gas phase. [4]

(c) Use your answers to (a) and (b) to explain why a temperature of 800 °C and a pressure of 30 atm are used for the manufacture of hydrogen from methane. [3]

(Total for Question 15 = 11 marks)